

CALIBRATION CERTIFICATE

Report Number: 859803

Sensor Model: CX-1050-SD-HT-1.4L	Serial Number: X105783
Sensor Type: Cernox Resistor	Sales Order: 106635
Sensor Excitation: see <i>Test Data</i> page of report	Date: November 06, 2015
Temperature Range: 1.40 K to 325 K	Due: November 05, 2016

Traceability and Calibration Method

This temperature sensor has been calibrated to the International Temperature Scale of 1990 (ITS-90) or the Provisional Low Temperature Scale (PLTS-2000) as appropriate. The calibrations are traceable to the National Institute of Standards and Technology (NIST, United States), the National Physical Laboratory (NPL, United Kingdom), the Physikalisch-Technische Bundesanstalt (PTB, Germany), or natural physical constants.

Lake Shore Cryotronics maintains ITS-90 and PLTS-2000 on standard platinum (PRT), rhodium-iron (RIRT), and germanium (GRT) resistance thermometers that have been calibrated directly by an internationally recognized national metrology institute (NIST, NPL, PTB) for $T < 330$ K or an ISO 17025 accredited metrology laboratory for 330 K $< T < 800$ K. A nuclear orientation thermometer is also used for temperatures less than 50 mK. These standards are routinely intercompared to verify consistency and accuracy of the temperature scale.

The sensor calibrations are performed by comparison to laboratory standard resistance thermometers and tested in accordance with Lake Shore Cryotronics, Inc. Quality Assurance Manual (QP-4220). The quality system of Lake Shore Cryotronics is registered to ISO 9001:2008.

Procedures used: 021-97-02, 099-00-00, 121-96-02, 029-95-02

Notes

The calibration results in this report apply only to the specific sensor specified above.

This report shall not be reproduced, except in full, without written approval from Lake Shore Cryotronics, Inc.

Unless stated otherwise, the uncertainties in this report are based on an approximate 95% confidence level with a coverage factor $k=2$.

Reported by: Matt Vance
Calibration Engineer/Technician

Approved by: John Krause
Metrology



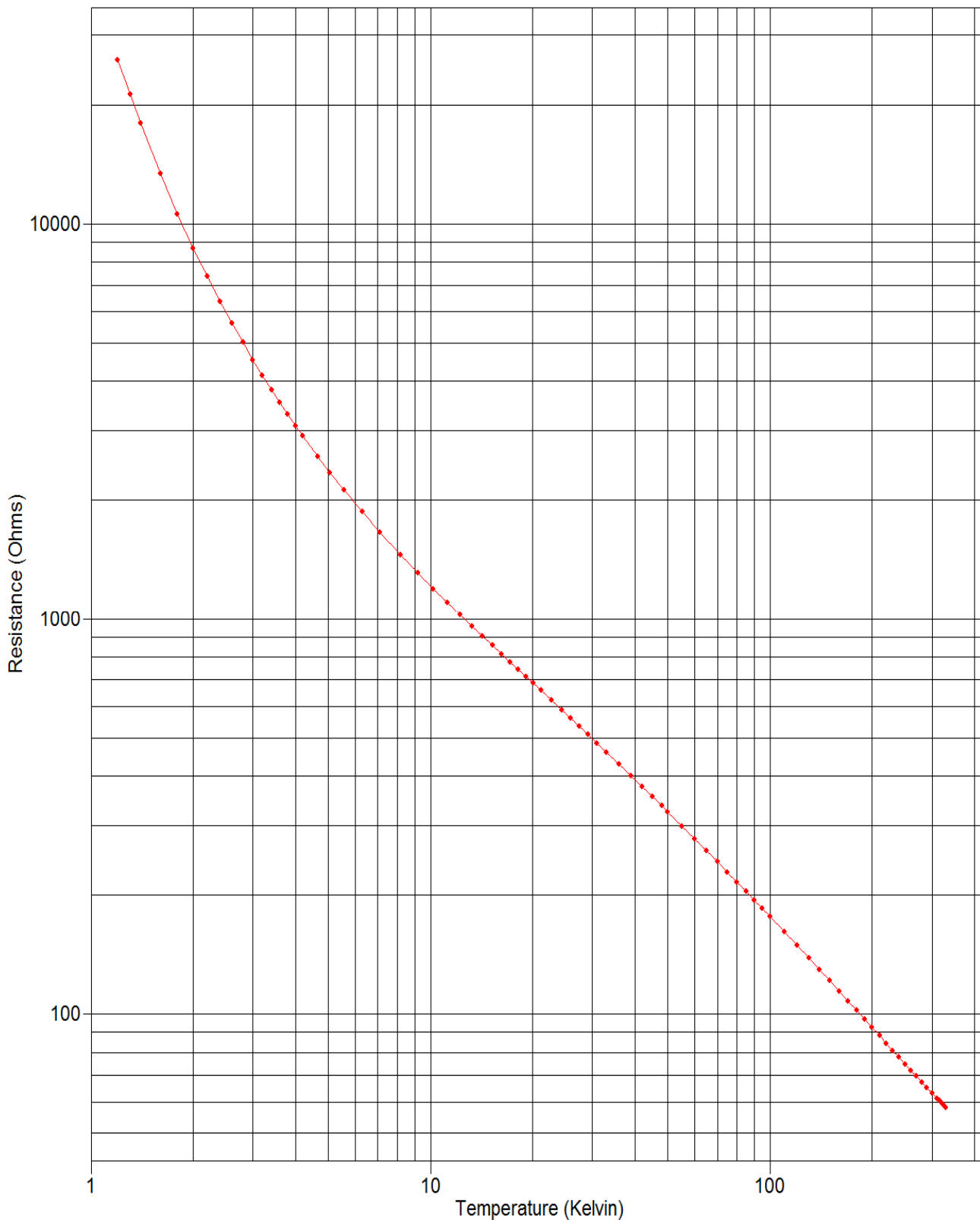
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F010-04-00_C

DATA PLOT

Calibration Report: 859803
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 106635
Serial Number: X105783
Temperature Range: 1.40 K to 325 K



TEST DATA

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Index	Temp. (K)	Resistance (Ω)	Excitation	Index	Temp. (K)	Resistance (Ω)	Excitation
1	1.19974	25988.0	2mV $\pm$ 25%	46	42.0053	375.952	2mV $\pm$ 25%
2	1.30361	21211.7	2mV $\pm$ 25%	47	45.0124	354.687	2mV $\pm$ 25%
3	1.40056	17939.1	2mV $\pm$ 25%	48	48.0102	335.854	2mV $\pm$ 25%
4	1.60208	13367.0	2mV $\pm$ 25%	49	50.0086	324.389	2mV $\pm$ 25%
5	1.79917	10589.0	2mV $\pm$ 25%	50	55.0008	298.999	2mV $\pm$ 25%
6	2.00126	8683.79	2mV $\pm$ 25%	51	60.0016	277.401	2mV $\pm$ 25%
7	2.20062	7352.91	2mV $\pm$ 25%	52	65.0033	258.776	2mV $\pm$ 25%
8	2.39789	6374.16	2mV $\pm$ 25%	53	70.0027	242.572	2mV $\pm$ 25%
9	2.59910	5608.04	2mV $\pm$ 25%	54	75.0029	228.299	2mV $\pm$ 25%
10	2.80024	5007.86	2mV $\pm$ 25%	55	80.0028	215.656	2mV $\pm$ 25%
11	3.00091	4527.25	2mV $\pm$ 25%	56	85.0008	204.353	2mV $\pm$ 25%
12	3.20068	4135.11	2mV $\pm$ 25%	57	90.0015	194.174	2mV $\pm$ 25%
13	3.40019	3809.55	2mV $\pm$ 25%	58	95.0062	184.960	2mV $\pm$ 25%
14	3.60023	3533.31	2mV $\pm$ 25%	59	100.000	176.598	2mV $\pm$ 25%
15	3.79957	3297.74	2mV $\pm$ 25%	60	110.002	161.916	2mV $\pm$ 25%
16	4.01020	3083.07	2mV $\pm$ 25%	61	120.007	149.476	2mV $\pm$ 25%
17	4.19679	2916.57	2mV $\pm$ 25%	62	130.003	138.803	2mV $\pm$ 25%
18	4.65344	2584.95	2mV $\pm$ 25%	63	140.021	129.511	2mV $\pm$ 25%
19	5.06107	2351.69	2mV $\pm$ 25%	64	150.032	121.377	2mV $\pm$ 25%
20	5.57283	2118.09	2mV $\pm$ 25%	65	160.032	114.211	2mV $\pm$ 25%
21	6.29121	1869.19	2mV $\pm$ 25%	66	170.040	107.828	2mV $\pm$ 25%
22	7.11179	1657.03	2mV $\pm$ 25%	67	180.048	102.123	2mV $\pm$ 25%
23	8.14207	1458.50	2mV $\pm$ 25%	68	190.056	97.0019	2mV $\pm$ 25%
24	9.17467	1309.24	2mV $\pm$ 25%	69	200.064	92.3774	2mV $\pm$ 25%
25	10.2053	1192.86	2mV $\pm$ 25%	70	210.066	88.1903	2mV $\pm$ 25%
26	11.2307	1099.33	2mV $\pm$ 25%	71	220.072	84.3758	2mV $\pm$ 25%
27	12.2436	1022.59	2mV $\pm$ 25%	72	230.087	80.8885	2mV $\pm$ 25%
28	13.2399	958.720	2mV $\pm$ 25%	73	240.097	77.7037	2mV $\pm$ 25%
29	14.2328	903.679	2mV $\pm$ 25%	74	250.092	74.7774	2mV $\pm$ 25%
30	15.2157	856.124	2mV $\pm$ 25%	75	260.086	72.0729	2mV $\pm$ 25%
31	16.1927	814.334	2mV $\pm$ 25%	76	270.110	69.5864	2mV $\pm$ 25%
32	17.1643	777.228	2mV $\pm$ 25%	77	280.115	67.2783	2mV $\pm$ 25%
33	18.1334	743.876	2mV $\pm$ 25%	78	290.108	65.1393	2mV $\pm$ 25%
34	19.1017	713.660	2mV $\pm$ 25%	79	300.129	63.1523	2mV $\pm$ 25%
35	20.0725	686.029	2mV $\pm$ 25%	80	310.119	61.3005	2mV $\pm$ 25%
36	21.1416	658.185	2mV $\pm$ 25%	81	315.125	60.4192	2mV $\pm$ 25%
37	22.7197	621.431	2mV $\pm$ 25%	82	320.122	59.5691	2mV $\pm$ 25%
38	24.3180	588.539	2mV $\pm$ 25%	83	326.153	58.5798	2mV $\pm$ 25%
39	25.9064	559.295	2mV $\pm$ 25%	84	330.128	57.9506	2mV $\pm$ 25%
40	27.5013	532.976	2mV $\pm$ 25%				
41	29.1123	509.004	2mV $\pm$ 25%				
42	30.9084	484.778	2mV $\pm$ 25%				
43	33.0095	459.364	2mV $\pm$ 25%				
44	36.0066	427.510	2mV $\pm$ 25%				
45	39.0067	400.001	2mV $\pm$ 25%				



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UNCERTAINTY ANALYSIS

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Calibration Data Uncertainty

The uncertainties of the measured calibration data for Lake Shore's sensors are summarized in the table below. The values given are the combined uncertainty of the temperature measurement and the resistance or voltage measurement expressed as an equivalent temperature uncertainty in millikelvin (mK). Note that the values are the calibration uncertainty only and do not include the stability of the temperature sensor. The uncertainty analysis has followed the guidelines for determining measurement uncertainty as outlined in the ISO Guide to the Expression of Uncertainty in Measurement, NIST Technical Note 1297, and ANSI/NCSL Z540-2-1997. Since the uncertainty varies with temperature due to the variation of the sensor sensitivity and excitation, the table gives typical values at several different temperatures throughout the range of the calibration. The uncertainty is based on an approximate 95% confidence level with a coverage factor $k = 2$.

T (K)	Uncertainty ($\pm$ mK)												
	GR	Cernox (CX)					RX			Platinum		RF-800	Diode
		1010	1030	1050	1070	1080	102A	103A	202A	100 Ω	25 Ω	27 Ω	
1.4	4	4	4	4			4	4	4			5	7
4.2	4	4	4	4	4		4	6	5			5	5
10	4	5	5	4	4		10	15	12			7	6
20	8	10	9	8	8	8	35	35	28	9	10	13	9
30	9	13	11	9	9	9	76	61	46	9	9	14	31
50	11	18	14	12	12	11				10	10	13	37
100	20	29	22	17	16	14				11	12	12	32
300		78	60	46	45	36				24	24	25	35
400		124	94	74	72	60				45	45	45	49
500										51	51		54

Polynomial Fit Uncertainty

When a sensor is used to measure temperature, a polynomial fit to the measured calibration data is often used to convert the sensor resistance (R) or voltage (V) to a temperature (T). How well the polynomial represents the sensor calibration data is another source of uncertainty when using the sensor. In the polynomials provided with this set of calibration data, the standard deviation of the fit can be used as an estimate of this additional temperature uncertainty. The standard deviation of fit is determined from the following equation:

$$\sigma_{fit}^2 = \frac{\sum_{i=1}^N (T_i - T_{icalc})^2}{N - n} = \frac{N}{N - n} (\Delta T_{RMS})^2$$

where

σ_{fit} = standard deviation of the fit

T_i = measured temperature for point i

T_{icalc} = the temperature calculated from the polynomial equation for point i

N = number of data points in fit range

n = number of fit coefficients

ΔT_{RMS} = root mean square deviation of fit

A value of ΔT_{RMS} is given for each range of fit.

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POLYNOMIAL EQUATION

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

1.40 K to 14.2 K
1.796e+4 ohms to 903.7 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 2.91080265441

ZU = 4.4147736053

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	5.502232	1.7737E-04	31021.05
1	-6.352481	2.8919E-04	-21966.29
2	2.851414	2.5848E-04	11031.64
3	-1.063305	2.6055E-04	-4080.94
4	0.331491	2.4703E-04	1341.89
5	-0.082360	2.2524E-04	-365.66
6	0.012775	2.1877E-04	58.39
7	0.000807	2.3452E-04	3.44
8	-0.001858	2.4117E-04	-7.70
9	0.001036	2.3171E-04	4.47

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i * \text{COS}(i * \text{ARCCOS}(k))$, where $0 \leq i \leq 9$
and the A_i 's are the coefficients in the table above.



POLYNOMIAL EQUATION

Calibration Report: 859803
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 106635
Serial Number: X105783
Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev
Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
1	25988.04	1.19974	1.19975	-0.01
2	21211.74	1.30361	1.30355	0.06
3	17939.11	1.40056	1.40067	-0.11
4	13367.00	1.60208	1.60193	0.16
5	10588.96	1.79917	1.79927	-0.10
6	8683.788	2.00126	2.00137	-0.11
7	7352.909	2.20062	2.20052	0.10
8	6374.163	2.39789	2.39770	0.19
9	5608.036	2.59910	2.59918	-0.09
10	5007.859	2.80024	2.80035	-0.11
11	4527.246	3.00091	3.00098	-0.07
12	4135.111	3.20068	3.20085	-0.17
13	3809.551	3.40019	3.40000	0.19
14	3533.312	3.60023	3.59999	0.25
15	3297.744	3.79957	3.79937	0.20
16	3083.071	4.01020	4.01031	-0.11
17	2916.571	4.19679	4.19769	-0.90
18	2584.948	4.65344	4.65232	1.12
19	2351.689	5.06107	5.06046	0.61
20	2118.094	5.57283	5.57466	-1.83
21	1869.189	6.29121	6.29161	-0.40
22	1657.028	7.11179	7.10977	2.02
23	1458.505	8.14207	8.14219	-0.12
24	1309.239	9.17467	9.17575	-1.08
25	1192.859	10.20534	10.20571	-0.37
26	1099.328	11.23066	11.23013	0.53
27	1022.587	12.24364	12.24371	-0.07
28	958.7201	13.23992	13.23887	1.05
29	903.6791	14.23283	14.23344	-0.61
30	856.1245	15.21570	15.21630	-0.60
31	814.3342	16.19275	16.19236	0.39

Order of Fit = 9 RMS error of fit = 0.67 mK
Largest absolute error = 2.02 mK at data point no. 22



POLYNOMIAL EQUATION

Calibration Report: 859803
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 106635
Serial Number: X105783
Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

14.2 K to 80.0 K
903.7 ohms to 215.7 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 2.28819161595 ZU = 3.00970018819

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	42.560275	4.0640E-04	104724.78
1	-37.771444	6.6287E-04	-56981.53
2	8.442821	6.0520E-04	13950.51
3	-1.101002	5.6872E-04	-1935.93
4	0.125008	5.4429E-04	229.67
5	-0.007075	5.1215E-04	-13.81
6	-0.005402	5.1306E-04	-10.53

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i * \text{COS}(i * \text{ARCCOS}(k))$, where $0 \leq i \leq 6$
and the A_i 's are the coefficients in the table above.



POLYNOMIAL EQUATION

Calibration Report: 859803
Sensor Model: CX-1050-SD-HT-1.4L
Sensor Type: Cernox Resistor

Sales Order: 106635
Serial Number: X105783
Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev
Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
27	1022.587	12.24371	12.24318	0.53
28	958.7201	13.23887	13.23951	-0.64
29	903.6791	14.23344	14.23372	-0.28
30	856.1245	15.21570	15.21631	-0.61
31	814.3342	16.19275	16.19275	-0.01
32	777.2281	17.16428	17.16363	0.65
33	743.8763	18.13339	18.13271	0.68
34	713.6599	19.10175	19.10090	0.84
35	686.0293	20.07246	20.07140	1.06
36	658.1849	21.14161	21.14247	-0.86
37	621.4311	22.71965	22.72053	-0.88
38	588.5395	24.31804	24.31743	0.61
39	559.2945	25.90637	25.90957	-3.20
40	532.9761	27.50132	27.50398	-2.67
41	509.0042	29.11230	29.11027	2.03
42	484.7785	30.90839	30.90573	2.67
43	459.3642	33.00951	33.00526	4.25
44	427.5096	36.00665	36.00929	-2.65
45	400.0013	39.00673	39.00845	-1.73
46	375.9525	42.00531	42.00731	-2.00
47	354.6874	45.01240	45.01320	-0.81
48	335.8537	48.01023	48.00703	3.19
49	324.3887	50.00861	50.00675	1.86
50	298.9994	55.00084	55.00307	-2.24
51	277.4012	60.00160	60.00104	0.57
52	258.7759	65.00326	65.00429	-1.03
53	242.5719	70.00272	70.00118	1.55
54	228.2991	75.00288	75.00510	-2.21
55	215.6561	80.00279	80.00188	0.91
56	204.3535	85.00078	84.99963	1.15
57	194.1742	90.00147	90.00222	-0.76

Order of Fit = 6 RMS error of fit = 1.77 mK
Largest absolute error = 4.25 mK at data point no. 43



POLYNOMIAL EQUATION

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

80.0 K to 325 K
215.7 ohms to 58.77 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 1.76305801043

ZU = 2.38484047587

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	176.265788	1.6050E-03	109820.42
1	-126.460380	2.4839E-03	-50912.39
2	23.105565	2.3863E-03	9682.53
3	-3.465637	2.2646E-03	-1530.33
4	0.673080	2.1565E-03	312.12
5	-0.135010	2.1527E-03	-62.72
6	0.019414	2.1172E-03	9.17

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i \cdot \text{COS}(i \cdot \text{ARCCOS}(k))$, where $0 \leq i \leq 6$
and the A_i 's are the coefficients in the table above.



POLYNOMIAL EQUATION

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Polynomial Type: Chebychev

Temp. (K) vs. Log(Resistance)

	R Meas. (Ω)	T Meas. (K)	T Eq. (K)	T diff. (mK)
53	242.5719	70.00118	70.00282	-1.64
54	228.2991	75.00510	75.00337	1.73
55	215.6561	80.00188	80.00049	1.40
56	204.3535	85.00078	84.99922	1.57
57	194.1742	90.00147	90.00278	-1.32
58	184.9601	95.00621	95.00766	-1.45
59	176.5980	100.00017	100.00073	-0.56
60	161.9165	110.00226	110.00652	-4.26
61	149.4763	120.00681	120.00716	-0.34
62	138.8032	130.00271	129.99747	5.23
63	129.5111	140.02101	140.01754	3.47
64	121.3775	150.03190	150.03185	0.05
65	114.2106	160.03153	160.02702	4.51
66	107.8282	170.03956	170.04185	-2.29
67	102.1226	180.04806	180.05721	-9.15
68	97.00191	190.05590	190.06018	-4.27
69	92.37742	200.06407	200.06684	-2.77
70	88.19027	210.06604	210.06153	4.51
71	84.37579	220.07156	220.06834	3.22
72	80.88853	230.08711	230.08979	-2.68
73	77.70366	240.09701	240.08371	13.30
74	74.77740	250.09222	250.07998	12.25
75	72.07289	260.08581	260.11175	-25.94
76	69.58635	270.10973	270.10056	9.17
77	67.27830	280.11535	280.11691	-1.56
78	65.13931	290.10752	290.12237	-14.85
79	63.15226	300.12859	300.11792	10.67
80	61.30045	310.11927	310.11406	5.21
81	60.41918	315.12494	315.12424	0.71
82	59.56914	320.12172	320.12305	-1.33
83	58.57978	326.15257	326.15858	-6.01
84	57.95061	330.12831	330.12488	3.44

Order of Fit = 6 RMS error of fit = 7.38 mK

Largest absolute error = -25.94 mK at data point no. 75



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INTERPOLATION TABLE

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>	<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>
1.400	17958.7	-29297	-2.2839	15.50	843.468	-43.835	-0.80553
1.500	15388.6	-22479	-2.1911	16.00	822.201	-41.276	-0.80322
1.600	13400.7	-17537	-2.0939	16.50	802.151	-38.962	-0.80144
1.700	11836.6	-13937	-2.0017	17.00	783.205	-36.854	-0.79994
1.800	10580.7	-11318	-1.9254	17.50	765.266	-34.930	-0.79877
1.900	9551.26	-9356.6	-1.8613	18.00	748.248	-33.167	-0.79787
2.000	8694.55	-7842.3	-1.8040	18.50	732.075	-31.549	-0.79727
2.100	7971.98	-6653.8	-1.7528	19.00	716.678	-30.058	-0.79687
2.200	7355.85	-5704.1	-1.7060	19.50	701.998	-28.682	-0.79671
2.300	6825.16	-4935.2	-1.6631	20.00	687.980	-27.406	-0.79672
2.400	6364.24	-4303.8	-1.6230	21.00	661.743	-25.123	-0.79725
2.500	5960.81	-3780.4	-1.5855	22.00	637.635	-23.137	-0.79828
2.600	5605.30	-3342.6	-1.5504	23.00	615.387	-21.398	-0.79974
2.700	5290.01	-2973.3	-1.5176	24.00	594.772	-19.862	-0.80146
2.800	5008.80	-2659.4	-1.4867	25.00	575.605	-18.499	-0.80344
2.900	4756.62	-2390.9	-1.4577	26.00	557.727	-17.280	-0.80556
3.000	4529.37	-2159.7	-1.4305	27.00	541.004	-16.186	-0.80778
3.100	4323.64	-1959.5	-1.4049	28.00	525.320	-15.198	-0.81009
3.200	4136.62	-1785.0	-1.3808	29.00	510.576	-14.304	-0.81242
3.300	3965.92	-1632.3	-1.3582	30.00	496.686	-13.490	-0.81477
3.400	3809.55	-1497.9	-1.3368	31.00	483.574	-12.746	-0.81711
3.500	3665.82	-1379.1	-1.3167	32.00	471.173	-12.065	-0.81943
3.600	3533.30	-1273.6	-1.2976	33.00	459.424	-11.440	-0.82173
3.700	3410.73	-1179.6	-1.2796	34.00	448.276	-10.864	-0.82398
3.800	3297.06	-1095.4	-1.2625	35.00	437.682	-10.331	-0.82613
3.900	3191.36	-1019.8	-1.2463	36.00	427.601	-9.8381	-0.82828
4.000	3092.85	-951.65	-1.2308	37.00	417.994	-9.3810	-0.83039
4.200	2914.64	-834.30	-1.2022	38.00	408.829	-8.9554	-0.83239
4.400	2757.79	-737.07	-1.1760	39.00	400.074	-8.5592	-0.83436
4.600	2618.75	-655.69	-1.1518	40.00	391.701	-8.1894	-0.83629
4.800	2494.64	-587.27	-1.1300	42.00	376.007	-7.5194	-0.83992
5.000	2383.18	-528.87	-1.1096	44.00	361.570	-6.9302	-0.84335
5.200	2282.53	-478.83	-1.0909	46.00	348.241	-6.4091	-0.84659
5.400	2191.20	-435.48	-1.0732	48.00	335.895	-5.9454	-0.84961
5.600	2107.95	-397.92	-1.0571	50.00	324.426	-5.5317	-0.85253
5.800	2031.73	-365.01	-1.0420	52.00	313.741	-5.1601	-0.85525
6.000	1961.69	-335.95	-1.0275	54.00	303.761	-4.8253	-0.85780
6.500	1809.13	-277.24	-0.99609	56.00	294.418	-4.5233	-0.86035
7.000	1682.12	-232.79	-0.96871	58.00	285.650	-4.2489	-0.86272
7.500	1574.63	-198.55	-0.94568	60.00	277.405	-3.9994	-0.86504
8.000	1482.40	-171.41	-0.92506	65.00	258.791	-3.4660	-0.87055
8.500	1402.29	-149.79	-0.90793	70.00	242.575	-3.0347	-0.87573
9.000	1331.97	-132.08	-0.89243	75.00	228.313	-2.6820	-0.88103
9.500	1269.68	-117.52	-0.87930	77.35	222.182	-2.5379	-0.88354
10.00	1214.06	-105.31	-0.86745	80.00	215.657	-2.3883	-0.88596
10.50	1164.04	-95.051	-0.85739	85.00	204.352	-2.1414	-0.89070
11.00	1118.76	-86.278	-0.84831	90.00	194.180	-1.9329	-0.89588
11.50	1077.55	-78.761	-0.84057	95.00	184.974	-1.7540	-0.90083
12.00	1039.84	-72.239	-0.83365	100.0	176.599	-1.5994	-0.90568
12.50	1005.17	-66.566	-0.82779	105.0	168.946	-1.4646	-0.91027
13.00	973.163	-61.565	-0.82241	110.0	161.925	-1.3464	-0.91465
13.50	943.507	-57.150	-0.81772	115.0	155.459	-1.2421	-0.91885
14.00	915.928	-53.248	-0.81390	120.0	149.485	-1.1495	-0.92280
14.50	890.184	-49.792	-0.81106	125.0	143.947	-1.0670	-0.92652
15.00	866.082	-46.667	-0.80825	130.0	138.801	-0.99296	-0.93000



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INTERPOLATION TABLE

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>	<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>
135.0	134.005	-0.92637	-0.93325	235.0	79.2894	-0.31878	-0.94482
140.0	129.526	-0.86620	-0.93624	240.0	77.7292	-0.30542	-0.94302
145.0	125.334	-0.81164	-0.93899	245.0	76.2339	-0.29282	-0.94105
150.0	121.402	-0.76198	-0.94148	250.0	74.7999	-0.28093	-0.93893
155.0	117.707	-0.71665	-0.94371	255.0	73.4236	-0.26970	-0.93667
160.0	114.229	-0.67515	-0.94568	260.0	72.1018	-0.25909	-0.93428
165.0	110.950	-0.63704	-0.94738	265.0	70.8317	-0.24905	-0.93176
170.0	107.853	-0.60196	-0.94881	270.0	69.6104	-0.23955	-0.92913
175.0	104.926	-0.56959	-0.94999	273.15	68.8649	-0.23382	-0.92743
180.0	102.153	-0.53965	-0.95089	275.0	68.4354	-0.23054	-0.92640
185.0	99.5255	-0.51190	-0.95153	280.0	67.3042	-0.22200	-0.92358
190.0	97.0312	-0.48613	-0.95192	285.0	66.2147	-0.21390	-0.92068
195.0	94.6611	-0.46216	-0.95205	290.0	65.1645	-0.20621	-0.91770
200.0	92.4068	-0.43982	-0.95193	295.0	64.1519	-0.19890	-0.91465
205.0	90.2604	-0.41897	-0.95157	300.0	63.1749	-0.19196	-0.91155
210.0	88.2148	-0.39948	-0.95097	305.0	62.2318	-0.18535	-0.90840
215.0	86.2636	-0.38123	-0.95016	310.0	61.3209	-0.17906	-0.90521
220.0	84.4007	-0.36412	-0.94912	315.0	60.4407	-0.17307	-0.90198
225.0	82.6206	-0.34806	-0.94788	320.0	59.5897	-0.16736	-0.89872
230.0	80.9184	-0.33298	-0.94644	325.0	58.7667	-0.16191	-0.89544



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THERMAL CYCLE TESTING

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

This sensor was tested for repeatability through rapid thermal cycles from room temperature into liquid helium. During this test, the following four lead resistance values were recorded:

Approximately 305 K:	62.2 Ω
Liquid Nitrogen:	222 Ω
Liquid Helium:	2910 Ω

The nitrogen and helium values were recorded in OPEN dewars, so precision comparisons with calibration values or other thermal cycle test values should not be made.

Recommended Operating Parameters:

For sensors calibrated by Lake Shore, the current to the sensor is adjusted to maintain the sensor output voltage or power at the values listed on the Test Data page.



BREAKPOINTS CUBIC SPLINE FORMAT

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Sensor Model: CX-1050-SD-HT-1.4L
Serial Number: X105783
Data Format: 7 (Ohms/Kelvin)
Setpoint Limit: 325

Measurement (ohms)	Temp (K)	Curvature	Measurement (ohms)	Temp (K)	Curvature
5.79506E+01	3.30125E+02	2.63396E-01	9.03679E+02	1.42334E+01	5.07287E-05
5.85798E+01	3.26159E+02	2.53525E-01	9.58720E+02	1.32389E+01	4.19274E-05
5.95691E+01	3.20123E+02	2.38002E-01	1.02259E+03	1.22437E+01	3.38821E-05
6.04192E+01	3.15124E+02	2.25890E-01	1.09933E+03	1.12301E+01	2.65710E-05
6.13005E+01	3.10114E+02	2.13877E-01	1.19286E+03	1.02057E+01	2.01003E-05
6.31523E+01	3.00118E+02	1.91519E-01	1.30924E+03	9.17575E+00	1.45520E-05
6.51393E+01	2.90122E+02	1.70943E-01	1.45850E+03	8.14219E+00	9.93687E-06
6.72783E+01	2.80117E+02	1.51945E-01	1.65703E+03	7.10977E+00	6.34177E-06
6.95864E+01	2.70101E+02	1.34505E-01	1.86919E+03	6.29161E+00	4.16420E-06
7.20729E+01	2.60112E+02	1.18611E-01	2.11809E+03	5.57466E+00	2.69159E-06
7.47774E+01	2.50080E+02	1.04089E-01	2.35169E+03	5.06046E+00	1.88886E-06
7.77037E+01	2.40084E+02	9.09847E-02	2.58495E+03	4.65232E+00	1.35741E-06
8.08885E+01	2.30090E+02	7.91678E-02	2.91657E+03	4.19769E+00	9.00342E-07
8.43758E+01	2.20068E+02	6.85330E-02	3.08307E+03	4.01031E+00	7.57922E-07
8.81903E+01	2.10062E+02	5.90475E-02	3.29774E+03	3.79937E+00	6.01500E-07
9.23774E+01	2.00067E+02	5.06209E-02	3.53331E+03	3.59999E+00	4.78713E-07
9.70019E+01	1.90060E+02	4.31518E-02	3.80955E+03	3.40000E+00	3.72647E-07
1.02123E+02	1.80057E+02	3.65714E-02	4.13511E+03	3.20085E+00	2.84387E-07
1.07828E+02	1.70042E+02	3.07896E-02	4.52725E+03	3.00098E+00	2.11262E-07
1.14211E+02	1.60027E+02	2.57384E-02	5.00786E+03	2.80035E+00	1.52080E-07
1.21377E+02	1.50032E+02	2.13502E-02	5.60804E+03	2.59918E+00	1.05465E-07
1.29511E+02	1.40018E+02	1.75388E-02	6.37416E+03	2.39770E+00	7.00821E-08
1.38803E+02	1.29997E+02	1.42480E-02	7.35291E+03	2.20052E+00	4.47532E-08
1.49476E+02	1.20007E+02	1.14324E-02	8.68379E+03	2.00137E+00	2.68075E-08
1.61916E+02	1.10007E+02	9.01945E-03	1.05890E+04	1.79927E+00	1.48243E-08
1.76598E+02	1.00001E+02	7.00163E-03	1.33670E+04	1.60193E+00	7.32020E-09
1.84960E+02	9.50077E+01	6.14599E-03	1.79391E+04	1.40067E+00	2.91354E-09
1.94174E+02	9.00028E+01	5.31640E-03	2.12117E+04	1.30355E+00	2.09900E-09
2.04353E+02	8.49992E+01	4.61099E-03	2.59880E+04	1.19975E+00	9.10189E-10
2.15656E+02	8.00019E+01	3.91295E-03			
2.28299E+02	7.50051E+01	3.32080E-03			
2.42572E+02	7.00012E+01	2.77098E-03			
2.58776E+02	6.50043E+01	2.28385E-03			
2.77401E+02	6.00010E+01	1.85173E-03			
2.98999E+02	5.50031E+01	1.47348E-03			
3.24389E+02	5.00067E+01	1.15153E-03			
3.35854E+02	4.80070E+01	1.03987E-03			
3.54687E+02	4.50132E+01	8.78180E-04			
3.75952E+02	4.20073E+01	7.35208E-04			
4.00001E+02	3.90085E+01	6.08072E-04			
4.27510E+02	3.60093E+01	4.96073E-04			
4.59364E+02	3.30053E+01	3.99102E-04			
4.84778E+02	3.09057E+01	3.39877E-04			
5.09004E+02	2.91103E+01	2.93384E-04			
5.32976E+02	2.75040E+01	2.55522E-04			
5.59295E+02	2.59096E+01	2.21063E-04			
5.88539E+02	2.43174E+01	1.89753E-04			
6.21431E+02	2.27205E+01	1.61196E-04			
6.58185E+02	2.11425E+01	1.35823E-04			
6.86029E+02	2.00714E+01	1.20115E-04			
7.13660E+02	1.91009E+01	1.06605E-04			
7.43876E+02	1.81327E+01	9.40020E-05			
7.77228E+02	1.71636E+01	8.21499E-05			
8.14334E+02	1.61928E+01	7.11211E-05			
8.56124E+02	1.52163E+01	6.04351E-05			



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BREAKPOINTS 340 FORMAT

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Name: CX-1050-SD-HT-1.4L
Serial number: X105783
Format: 4 ;Log Ohms/Kelvin
Limit: 325.0
Coefficient: 1 ;Negative

Point 1: 1.76911,325.000	Point 51: 2.27541, 93.000	Point 101: 2.95930, 14.100
Point 2: 1.77638,319.000	Point 52: 2.28603, 90.500	Point 102: 2.97338, 13.550
Point 3: 1.78319,313.500	Point 53: 2.29692, 88.000	Point 103: 2.98677, 13.050
Point 4: 1.79014,308.000	Point 54: 2.30809, 85.500	Point 104: 3.00075, 12.550
Point 5: 1.79725,302.500	Point 55: 2.31956, 83.000	Point 105: 3.01541, 12.050
Point 6: 1.80452,297.000	Point 56: 2.33134, 80.500	Point 106: 3.03079, 11.550
Point 7: 1.81194,291.500	Point 57: 2.34347, 78.000	Point 107: 3.04700, 11.050
Point 8: 1.81954,286.000	Point 58: 2.35596, 75.500	Point 108: 3.06239, 10.600
Point 9: 1.82731,280.500	Point 59: 2.36624, 73.500	Point 109: 3.07857, 10.150
Point 10: 1.83527,275.000	Point 60: 2.37676, 71.500	Point 110: 3.09569, 9.700
Point 11: 1.84341,269.500	Point 61: 2.38756, 69.500	Point 111: 3.11179, 9.300
Point 12: 1.85174,264.000	Point 62: 2.39864, 67.500	Point 112: 3.12876, 8.900
Point 13: 1.86028,258.500	Point 63: 2.41003, 65.500	Point 113: 3.14676, 8.500
Point 14: 1.86822,253.500	Point 64: 2.42175, 63.500	Point 114: 3.16589, 8.100
Point 15: 1.87634,248.500	Point 65: 2.43381, 61.500	Point 115: 3.18631, 7.700
Point 16: 1.88465,243.500	Point 66: 2.44686, 59.400	Point 116: 3.20543, 7.350
Point 17: 1.89314,238.500	Point 67: 2.45970, 57.400	Point 117: 3.22576, 7.000
Point 18: 1.90183,233.500	Point 68: 2.47295, 55.400	Point 118: 3.24753, 6.650
Point 19: 1.91072,228.500	Point 69: 2.48528, 53.600	Point 119: 3.27094, 6.300
Point 20: 1.91983,223.500	Point 70: 2.49797, 51.800	Point 120: 3.29699, 5.940
Point 21: 1.92915,218.500	Point 71: 2.51108, 50.000	Point 121: 3.32206, 5.620
Point 22: 1.93870,213.500	Point 72: 2.52464, 48.200	Point 122: 3.34751, 5.320
Point 23: 1.94848,208.500	Point 73: 2.53866, 46.400	Point 123: 3.37504, 5.020
Point 24: 1.95851,203.500	Point 74: 2.55319, 44.600	Point 124: 3.40302, 4.740
Point 25: 1.96880,198.500	Point 75: 2.56657, 43.000	Point 125: 3.43344, 4.460
Point 26: 1.97934,193.500	Point 76: 2.58040, 41.400	Point 126: 3.46438, 4.200
Point 27: 1.99017,188.500	Point 77: 2.59565, 39.700	Point 127: 3.49157, 3.990
Point 28: 2.00128,183.500	Point 78: 2.61055, 38.100	Point 128: 3.51654, 3.810
Point 29: 2.01154,179.000	Point 79: 2.62603, 36.500	Point 129: 3.54184, 3.640
Point 30: 2.02204,174.500	Point 80: 2.64112, 35.000	Point 130: 3.56894, 3.470
Point 31: 2.03281,170.000	Point 81: 2.65679, 33.500	Point 131: 3.59641, 3.310
Point 32: 2.04386,165.500	Point 82: 2.67313, 32.000	Point 132: 3.62594, 3.150
Point 33: 2.05519,161.000	Point 83: 2.68902, 30.600	Point 133: 3.65587, 3.000
Point 34: 2.06683,156.500	Point 84: 2.70558, 29.200	Point 134: 3.68815, 2.850
Point 35: 2.07878,152.000	Point 85: 2.72288, 27.800	Point 135: 3.72322, 2.700
Point 36: 2.09107,147.500	Point 86: 2.73970, 26.500	Point 136: 3.75886, 2.560
Point 37: 2.10370,143.000	Point 87: 2.75728, 25.200	Point 137: 3.79763, 2.420
Point 38: 2.11671,138.500	Point 88: 2.77573, 23.900	Point 138: 3.83701, 2.290
Point 39: 2.12860,134.500	Point 89: 2.79364, 22.700	Point 139: 3.87997, 2.160
Point 40: 2.14081,130.500	Point 90: 2.81246, 21.500	Point 140: 3.92726, 2.030
Point 41: 2.15337,126.500	Point 91: 2.83066, 20.400	Point 141: 3.97547, 1.910
Point 42: 2.16630,122.500	Point 92: 2.84453, 19.600	Point 142: 4.02872, 1.790
Point 43: 2.17960,118.500	Point 93: 2.85621, 18.950	Point 143: 4.08312, 1.680
Point 44: 2.19332,114.500	Point 94: 2.86829, 18.300	Point 144: 4.13817, 1.580
Point 45: 2.20748,110.500	Point 95: 2.88082, 17.650	Point 145: 4.19949, 1.480
Point 46: 2.22210,106.500	Point 96: 2.89383, 17.000	Point 146: 4.25400, 1.400
Point 47: 2.23532,103.000	Point 97: 2.90634, 16.400	
Point 48: 2.24502,100.500	Point 98: 2.91933, 15.800	
Point 49: 2.25491, 98.000	Point 99: 2.93287, 15.200	
Point 50: 2.26504, 95.500	Point 100: 2.94582, 14.650	



BREAKPOINTS 91C/93C/330 FORMAT

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Interpolation Method: Lagrangian

Limit: 325.0 (Kelvin)

Format: 4 (Log Ohms/Kelvin)

Number of Breakpoints: 52

No.	Units	Temperature (K)	No.	Units	Temperature (K)
1	1.76913	325.0	31	3.05211	10.9
2	1.77033	324.0	32	3.11184	9.3
3	1.78888	309.0	33	3.17097	8.0
4	1.80856	294.0	34	3.22586	7.0
5	1.82948	279.0	35	3.28528	6.1
6	1.85176	264.0	36	3.34068	5.4
7	1.87554	249.0	37	3.39701	4.8
8	1.90097	234.0	38	3.45237	4.3
9	1.92822	219.0	39	3.50398	3.9
10	1.95752	204.0	40	3.56417	3.5
11	1.98909	189.0	41	3.61665	3.2
12	2.02325	174.0	42	3.67730	2.9
13	2.06035	159.0	43	3.72346	2.7
14	2.10089	144.0	44	3.77530	2.5
15	2.14551	129.0	45	3.83411	2.3
16	2.19510	114.0	46	3.86663	2.2
17	2.25094	99.0	47	3.93925	2.0
18	2.31495	84.0	48	3.98006	1.9
19	2.39032	69.0	49	4.02452	1.8
20	2.44626	59.5	50	4.07323	1.7
21	2.51112	50.0	51	4.18720	1.5
22	2.54995	45.0	52	4.25428	1.4
23	2.59296	40.0			
24	2.64116	35.0			
25	2.69608	30.0			
26	2.75873	25.1			
27	2.81741	21.2			
28	2.87792	17.8			
29	2.93523	15.1			
30	2.99373	12.8			

Temperature for Resistance Decades:

Res. (Ohms)	Temp. (K)
100	184.076
1000	12.577
10000	1.854



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BREAKPOINTS 234 FORMAT

Calibration Report: 859803

Sensor Model: CX-1050-SD-HT-1.4L

Sensor Type: Cernox Resistor

Sales Order: 106635

Serial Number: X105783

Temperature Range: 1.40 K to 325 K

Maximum Temperature Error:

1.4 - 10 K:	0.010 K
10 - 20 K:	0.017 K
20 - 40 K:	0.008 K
40 - 100 K:	0.017 K
> 100 K:	0.073 K

BP #	Temp. (K)	Res. (Ω)	Log10 Res.	BP #	Temp. (K)	Res. (Ω)	Log10 Res.
1	316.071	60.25596	1.780	51	23.613	602.5596	2.780
2	300.413	63.09573	1.800	52	22.292	630.9573	2.800
3	285.681	66.06934	1.820	53	21.042	660.6934	2.820
4	271.796	69.18310	1.840	54	19.860	691.8310	2.840
5	258.688	72.44360	1.860	55	18.745	724.4360	2.860
6	246.292	75.85776	1.880	56	17.693	758.5776	2.880
7	234.551	79.43282	1.900	57	16.703	794.3282	2.900
8	223.415	83.17638	1.920	58	15.771	831.7638	2.920
9	212.837	87.09636	1.940	59	14.896	870.9636	2.940
10	202.779	91.20108	1.960	60	14.074	912.0108	2.960
11	193.203	95.49926	1.980	61	13.302	954.9926	2.980
12	184.077	100.0000	2.000	62	12.578	1000.000	3.000
13	175.374	104.7129	2.020	63	11.265	1096.478	3.040
14	167.068	109.6478	2.040	64	10.113	1202.264	3.080
15	159.136	114.8154	2.060	65	9.105	1318.257	3.120
16	151.557	120.2264	2.080	66	8.222	1445.440	3.160
17	144.315	125.8925	2.100	67	7.449	1584.893	3.200
18	137.391	131.8257	2.120	68	6.770	1737.801	3.240
19	130.772	138.0384	2.140	69	6.173	1905.461	3.280
20	124.443	144.5440	2.160	70	5.647	2089.296	3.320
21	118.392	151.3561	2.180	71	5.183	2290.868	3.360
22	112.607	158.4893	2.200	72	4.771	2511.886	3.400
23	107.076	165.9587	2.220	73	4.405	2754.229	3.440
24	101.791	173.7801	2.240	74	4.079	3019.952	3.480
25	96.741	181.9701	2.260	75	3.787	3311.311	3.520
26	91.916	190.5461	2.280	76	3.526	3630.781	3.560
27	87.308	199.5262	2.300	77	3.291	3981.072	3.600
28	82.909	208.9296	2.320	78	3.079	4365.158	3.640
29	78.713	218.7762	2.340	79	2.888	4786.301	3.680
30	74.712	229.0868	2.360	80	2.714	5248.075	3.720
31	70.897	239.8833	2.380	81	2.557	5754.399	3.760
32	67.262	251.1886	2.400	82	2.413	6309.573	3.800
33	63.798	263.0268	2.420	83	2.281	6918.310	3.840
34	60.499	275.4229	2.440	84	2.161	7585.776	3.880
35	57.358	288.4032	2.460	85	2.050	8317.638	3.920
36	54.368	301.9952	2.480	86	1.948	9120.108	3.960
37	51.522	316.2278	2.500	87	1.854	10000.00	4.000
38	48.813	331.1311	2.520	88	1.649	12589.25	4.100
39	46.236	346.7369	2.540	89	1.480	15848.93	4.200
40	43.783	363.0781	2.560	90	1.338	19952.62	4.300
41	41.450	380.1894	2.580	91	1.216	25118.86	4.400
42	39.231	398.1072	2.600				
43	37.120	416.8694	2.620				
44	35.113	436.5158	2.640				
45	33.205	457.0882	2.660				
46	31.392	478.6301	2.680				
47	29.669	501.1872	2.700				
48	28.034	524.8075	2.720				
49	26.481	549.5409	2.740				
50	25.009	575.4399	2.760				



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